MILITARY SPECIFICATION

CHEST: 10548700

1. SCOPE

1.1 This specification covers the performance requirements and quality assurance provisions for one type of reusable chest for Plotting Board M18.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-I-45607	Inspection Equipment, Acquisition, Disposition and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10548700	Chest, 10548700

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(Copies of specifications, standards and drawings required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The chest shall be manufactured in accordance with Drawing F10548700 and all drawings pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part identification and marking
(d) Workmanship

3.4 Environmental.

3.4.1 Storage.- The latched chest assembly shall show no evidence of distoration, cracks or delamination after having been thermally stabilized at -70°F and 160°F.

3.4.2 Standard ambient temperature.- Subsequent to meeting the requirement of 3.4.1, the chest shall meet all the requirements of 3.4.4, 3.4.5 and 3.5 at a standard ambient temperature between 60°F to 90°F.

3.4.3 Extreme operating temperature.- The chest shall meet all the requirements of 3.5.1 and 3.5.2 while thermally stabilized at -40°F and +150°F.

3.4.4 Impact resistance.- There shall be no evidence of loose or damaged parts to either the chest or plotting board as a result of the chest being dropped on each of the 8 corners and 6 faces from a height of 12 inches ± 2 inches onto a 2 inch ± 1/2 inch thick concrete surface. The chest assembly shall not become unlatched as a result of the impacts and shall meet the requirements of 3.4.5 and 3.5.

3.4.5 Watertightness.- The latched chest assembly with the plotting board installed shall show no evidence of water penetration after being submerged in 16 ± 2 inches of water for 5 ± 1 minute. The functioning portion of the breather valves shall be covered during the watertightness test.

3.5 Performance.

3.5.1 Fasteners.- All catches, hinges, straps, etc. shall function without interference. The external catches, when engaged, shall secure

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the cover to the body and shall displace a uniform compression on the gasketed seal.

3.5.2 Handles.- All handles shall function without interference and shall support the latched chest assembly with the M18 Plotting Board installed. There shall be no evidence of damage to, or separation of the handle assembly, and the supported load shall cause no damage to the cover or body in the vicinity of the handle attachments or elsewhere.

3.5.3 Interface requirements.- The chest shall interface, support, and retain the M18 Plotting Board. There shall be no free movement of any of the components after the plotting board components are properly arranged and secured within the chest and the cover is closed and latched to the body.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 chest assemblies manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I

CLASSIFICATION	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Fasteners	3.5.1	4.6.3
102. Handles	3.5.2	4.6.4
103. Interface requirements	3.5.3	4.6.7
MINOR: NONE		

NOTE: The tests in Table I shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling.

4.4.1 General.- Three chests shall be selected at random as a special sampling from each 100 produced. The samples shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
301.	Materials	3.1	Per applicable drawings and specifications
302.	Fabrication	3.2	Drawings - Visual
303.	General specification	3.3	MIL-F-13926 - Visual

NOTE: The tests in Table II shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.4.2 Environmental tests.- Three chests shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperatures	3.4.1	4.6.1
305.	Standard ambient temperature (60°F to 90°F)	3.4.2	4.6.2
306.	Extreme temperature operation (-40°F and 150°F)	3.4.3	4.6.2
307.	Impact resistance	3.4.4	4.6.5
308.	Watertightness	3.4.5	4.6.6

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The chest assemblies shall be exposed to the temperature conditions specified in 3.4.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The chests shall be placed in the test chamber and the temperature of the chamber reduced gradually to -70°F and shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period, the temperature shall then be gradually raised to +160°F and the chests shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) at which time the chests shall be removed from the test chamber. The chests shall then be subjected to a visual and tactile examination to determine that there is no evidence of distortion, cracks or delamination externally or internally to determine compliance with 3.4.1 and shall then be tested in accordance with 4.6.3 through 4.6.7 to determine compliance with 3.4.2.

NOTE: The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperature.- The chests shall be exposed to the temperature conditions specified in 3.4.3, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The chests shall be placed in the test chamber and the temperature of the chamber reduced gradually to -40°F. The chests shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period,

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and while at -40°F, the chests shall be subjected to the tests in 4.6.3 and 4.6.4 to determine compliance with 3.4.3. Upon completion of this test, the temperature of the test chamber shall be raised gradually to +150°F. The chests shall remain at this temperature for a period of 4 hours. At the completion of this 4 hour period, and while at +150°F, the chests shall again be subjected to the tests in 4.6.3 and 4.6.4 to determine compliance with 3.4.3. Upon completion of this test, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) at which time the chests shall be removed from the test chamber.

4.6.3 Fasteners.- Performance of the fastening devices shall be checked by a functional and visual examination when engaged and disengaged a minimum of 5 times. The supporting and securing devices provided in the interior of the chest shall be accessible and receive and secure the associated item without interference. The cover and body shall close without interference and shall secure the cover to the body as required. Upon completion of this test, with the chest closed, the lip edges of the body and cover shall be examined to determine uniformity under compression for compliance with 3.5.1.

4.6.4 Handles.- Performance of the handles shall be checked manually. The handles shall support the chest assembly when carried with its specified contents and shall be visually examined in the area of the handle attachments for damage for compliance with the requirements of 3.5.2.

4.6.5 Impact resistance.- This test shall be performed as specified in MIL-T-21200 (Drop test) with the following exceptions: The chest with the plotting board components properly secured shall be dropped as specified in 3.4.4. Subsequent to this test, the chest shall be opened and the chest and plotting board components shall be inspected by visual and tactile means for evidence of damage as specified in 3.4.4.

4.6.6 Watertightness.- This test shall be performed as specified in MIL-STD-810, Method 512, Procedure I with the following exceptions: The latched chest assembly with components installed shall be submerged as specified in 3.4.5. The valves shall be covered with waterproof tape prior to immersion. Subsequent to this test, the chest assembly shall be opened and visually inspected for evidence of water penetration as specified in 3.4.5.

4.6.7 Interface requirements.- The plotting board components shall be properly positioned in the spaces provided. The plotting board components shall then be manually manipulated to assure that there is no excess movement of the plotting board components in the allotted spaces. The cover shall then be closed and latched to the body utilizing normal hand pressure. The chest shall then be opened and inspected by visual and tactile means to assure that the plotting board components have not been displaced.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- Unless otherwise specified, the chest shall be prepared for shipment using normal commercial methods and materials to assure acceptance by common carrier or other carrier for safe transportation.

5.2 Marking.- Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The chest assembly defined herein is intended to house and transport Plotting Board M18. The chest consists of 3 major components; the body, the platform, and the cover. The body contains the platform and space for packing all components of the plotting board.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Applicable stock number
c. Provisions for first article testing.

Custodian:

Army - MU

Preparing activity:

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Project No. 1220-A260

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